CHALLENGE — REGISTRATION OPEN

Video to Data (V2D) Challenge

From Human Video to Grounded Data

Where does the pipeline from human video to grounded data break? Three coupled tracks answer this with shared task suites: 4D reconstruction, robotic grounding, and end-to-end egocentric transfer.

Register Now | Tracks | Awards | Judges | Timeline | Submission | Leaderboard | More Than a Leaderboard | Starter Toolkit

Watch Livestream Recording ↗ | CoRL Tutorial ↗

Why This, Why Now

Human video has the potential to be the largest and most scalable training corpus for robotics; every ingredient needed to use it is ready at once.

READY NOW

01 VLA and WAM are scaling beyond what teleoperation data can support.

02 4D human-object reconstruction can now produce digital twins of the real world for robots to learn within.

03 The advancements in humanoid and dexterous-hand hardware have narrowed the morphology gap between human demonstrations and deployable robots.

04 With massively parallel GPU simulation, human demonstrations can be transformed into deployable robotic skills through RL and generate diverse, physically grounded data at scale for model pre- and post-training.

STILL MISSING

What is still missing is *measurement* and *scale*.

Measurement, because existing benchmarks evaluate stages in isolation. Without evaluating every stage on the same tasks end-to-end, the field cannot localize where a pipeline breaks, trace how errors propagate, or quantify the downstream value of improving each component.

Scale, because success on a small, curated benchmark does not show whether a method will remain robust across thousands of tasks, objects, behaviors, and environments.

THIS CHALLENGE

This challenge supplies both. For measurement, we provide a representative benchmark that evaluates individual stages and the end-to-end pipeline under shared tasks, spanning both third-person and egocentric human video. For scale, participants develop and validate their solutions on a curated subset of our large-scale in-house dataset with held-out ground truth, supported by an end-to-end baseline. After the challenge, we evaluate the top solutions on the broader dataset to determine how well they generalize and perform at large scale.

Challenge Tracks

TRACK 01

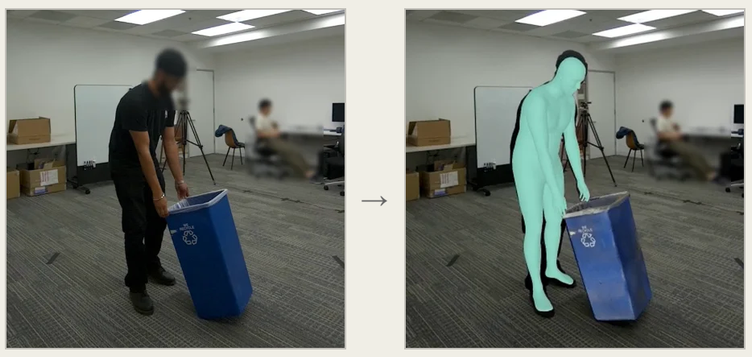
Reconstruction

Third-Person-View Video → 4D HOI

Can we reconstruct 4D human-object interaction from monocular video?

READ THE FULL TRACK

Challenge Dataset: track 1 ↗



THIRD-PERSON-VIEW VIDEO → 4D HOI

TRACK 02

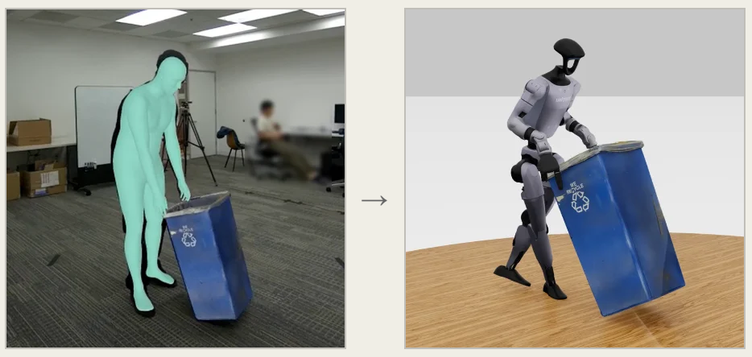
Robotic Grounding

Third-Person-View 4D HOI → Policy

How much reconstruction error can be tolerated?

READ THE FULL TRACK

Challenge Dataset: track 2 ↗



THIRD-PERSON-VIEW 4D HOI → POLICY

TRACK 03

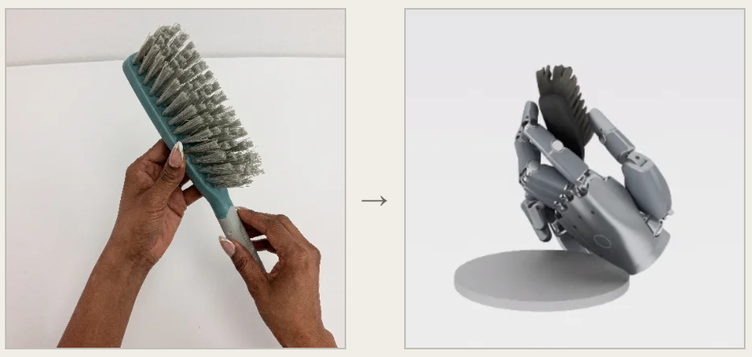
Egocentric

Egocentric Video → Policy

Can we holistically reconstruct
and transfer from egocentric video?

READ THE FULL TRACK

Challenge Dataset: track 3 ↗



EGOCENTRIC VIDEO → POLICY

Awards

Three winners per track will be announced at the CoRL 2026 tutorial, with a total prize pool of **$54K** across all tracks*. Two winners per track will be selected based on leaderboard rankings:

Track Winner: first on the track leaderboard.

`Prize: DGX Spark ($6K value)`

Track Runner-Up: second by the same metric.

`Prize: $3K GPU credits`

The **Industry Innovation Award** asks a different question: *Which solution would a robotics team actually put into production?*

Any team with a valid final submission, a commitment to release its code, and a technical report in any track is eligible, regardless of leaderboard rank. The invited panel will evaluate submissions against these criteria:

Economic Viability: Does it reduce the cost of simulation or data generation at scale?

Data Flywheel: Does it improve the system over time without additional manual effort?

Innovation: Does this challenge an assumption the field treats as fixed?

Complexity Reduction: Does it simplify the problem, reduce input-capture effort, and help another team deploy it faster?

`Prize: DGX Spark + $3K GPU credits ($9K value)`

Eligibility for all awards: a valid final submission, agreed code release (research license), and the short technical report. Organizer entries are non-competing.

Top teams will be invited to present at our CoRL 2026 tutorial, with remote participation available, and selected solutions will receive large-scale evaluation and upstream integration. Awardees will serve as core authors of the post-challenge technical report, while all teams with valid final submissions will be invited to contribute as co-authors.

*Terms and conditions apply

Judges



Hao Su
Fudan University
Sudo.ai
Website ↗



Yunzhu Li
Columbia University
SceniX & World Labs
Website ↗ LinkedIn ↗



Louis Lian
Lightwheel
LinkedIn ↗



Changliu Liu
CMU
Website ↗



Yuting Ye
Meta
Website ↗



Kaifeng Zhang
Sharpa
Website ↗



Javier Romero
Meta
Website ↗

Timeline

September 21	Registration
September 21 — Now	Challenge data release and official launch Challenge Dataset ↗
November 4, 2026 (Wed) 5:00 p.m. EST	Leaderboard freeze
November 12	Winner announced at our CoRL 2026 tutorial and invited to give a brief presentation to the community.
After November 12	Winner solution eval and tech report preparation

Register Now Teams may enter any subset of the three tracks.

Submission

We use Kaggle to process and score leaderboard submissions, and each metric has its own Kaggle leaderboard. Submissions across all tracks are displayed below. The easiest way to submit a solution is to use the agent skill included in the submission kit. Your agent will walk you through:

1. Creating a Kaggle account and generating a Kaggle API token.
2. Accepting the competition rules for each Kaggle competition.
3. Preparing your submission materials for the relevant track.
4. Running the evaluation, packing, and submission scripts, and seeing your results.

Each team may submit up to five times per week, with unlimited submissions during the final three days. For more information about submission instructions, required formats, and metrics, please see the submission kit. For Track 2 participants, please create a separate submission for each tier. Your final score is a weighted combination of the three tiers. For further questions about the submission process or scoring, see our FAQ Document.

Challenge Dataset ↗ | Submission Kit ↓

We ask that participants include a GitHub commit in their submissions, and share their code repository with Nvidia-V2D-Challenge.

Final submission containers. Teams submit a container alongside their technical report that reproduces their best or selected leaderboard submissions. Please refer to the submission kit for instructions and an agent skill to build the final container. Containers can be shared via Google Drive or another file sharing service. All leaderboard results are provisional pending container checks, and mismatches between a container's code, hashes, or results and those on the leaderboard can delay or invalidate prizes.

Leaderboard

The leaderboard is live! Submissions end on Wednesday, November 4, 2026, at 5:00 p.m. EST. **Standings here may take up to 24 hours to reflect a new submission**; your score appears on the Kaggle competition page as soon as it is computed. Each tier in Track 2 uses a different input reference trajectory (clean, noised, or reconstructed) and is measured against the same clean trajectory.

Click a metric to sort the leaderboard by that metric.

Track 1 | Track 2 · Tier 1 | Track 2 · Tier 2 | Track 2 · Tier 3 | Track 3

Track 1 - Reconstruction

#	TEAM	CD-H (CM) ▲	CD-O (CM)	ACC-H (CM)	ACC-O (CM)	PEN (CM)	LAST ENTRY
1	**xpz666** xpz666	**11.3702**	16.2895	0.1341	0.0703	0.0000	2026-11-04
2	**Rice RobotPI Lab** pchanrung, vectorwangel	**11.8310**	15.4545	0.1338	0.0682	0.0000	2026-11-04
3	**mparrrk** mparrrk	**12.3819**	16.0174	0.1360	0.0703	0.0002	2026-11-04
4	**keti4cv** shpark97	**12.3893**	15.7023	0.1342	0.0693	0.0000	2026-11-04
5	**Hyletic AI (lockpicker)** lockpicker	**12.4358**	16.2686	0.1319	0.0669	0.0000	2026-11-04
6	**lzus_systems** ibrahimmemon1811, swithinraj, ujjwalpatil7, zaid3727	**12.8344**	16.1514	0.1361	0.0697	0.0000	2026-11-04
7	**Lattice Studios** latticestudios	**12.9126**	17.5942	0.1331	0.0689	0.0015	2026-11-04
8	**BarcelonaHoneyCod** barcelonahoneycod	**12.9132**	17.2359	0.1350	0.0681	0.0000	2026-11-04
9	**EviisKitson** eviiskitson	**12.9487**	16.1884	0.1359	0.0697	0.0000	2026-11-02
10	**Chivatam** chvsthop, hannguyenngocbao	**12.9965**	18.4559	0.1363	0.0698	0.0000	2026-11-04
11	**Lodestar** lodestarmax	**13.2654**	17.5837	0.1352	0.0707	0.0000	2026-11-03
12	**Nickzz** nikhil125, sv1105	**13.3250**	18.1535	0.1381	0.0705	0.0000	2026-11-04
13	**namul2hand** namul2s	**13.5531**	24.4684	0.1357	0.0718	0.0006	2026-10-23
14	**Shubham** shubhamtyagi542	**13.6834**	18.6609	0.1375	0.0699	0.0000	2026-11-04
15	**Xueyi Wang (FF)** wxy821018	**13.6858**	19.5379	0.1354	0.0706	0.0000	2026-11-04
16	**SNU CVLAB** hyeongjinnam	**13.7059**	16.3460	0.1377	0.0743	0.0000	2026-11-04
17	**MassimilianoGhiotto** massimilianoghiotto	**13.7329**	24.4392	0.1354	0.0736	0.0000	2026-11-04
18	**domoarigato** cee123, tianchenwang816, yufeizhang96	**13.8010**	17.9403	0.1352	0.0703	0.0000	2026-11-04
19	**ByeByeHelloHaHaHa** byebyehellohahaha	**14.2040**	24.4783	0.1374	0.0732	0.0002	2026-11-04
20	**JaeyongPark** jaeyongpark0121	**14.3026**	71.3682	0.1732	0.2860	0.1821	2026-10-30
21	**QY Lin** doreen071	**14.5343**	22.2511	0.1374	0.0722	0.0000	2026-11-02
22	**NVSimranPatil** nvsimranpatil	**14.5804**	69.2677	0.1464	0.0743	0.0000	2026-11-04
23	**Mochii** brandonhsu111, hiveyuan, melanieww	**14.6207**	19.2654	0.1369	0.0688	0.0000	2026-11-04
24	**Debajyoti Dasgupta** debajyoti123	**14.6427**	24.4510	0.1379	0.0735	0.0000	2026-11-03
25	**Open4D** alex0807, gsadhas	**14.6781**	18.5737	0.1375	0.0702	0.0000	2026-11-04
26	**Zesen Zhao** zesenzhao	**14.7430**	17.3482	0.1422	0.0699	0.0013	2026-11-04
27	**Pragun Sharma** pgsharma	**14.8124**	31.5362	0.1390	0.0771	0.0000	2026-11-04
28	**unlimited-4D** jixianricardowu, thuqixuan	**14.8249**	18.0742	0.1402	0.0703	0.0000	2026-11-04
29	**Ankita Bajaj** nvankitabajaj	**14.8425**	26.2437	0.1784	0.1129	0.0000	2026-10-25
30	**Lab2A** vihaanagarwal123	**14.8446**	29.9685	0.1396	0.0729	0.0000	2026-11-04
31	**DAInamite** xuyuan	**14.8552**	22.2941	1.2977	1.0392	0.0013	2026-11-01
32	**um** chahyonku	**14.8574**	26.2605	0.1719	0.1221	0.0702	2026-11-02

#	TEAM	CD-H (CM) ▲	CD-O (CM)	ACC-H (CM)	ACC-O (CM)	PEN (CM)	LAST ENTRY
33	Neural Motion robertzty54	**14.8856**	42.9573	0.1394	0.0796	0.0000	2026-11-04
34	quixoteh quixoteh	**14.9138**	67.8452	0.4827	0.0745	0.0023	2026-11-04
35	Rock Paper Grippers surtantheta	**14.9953**	29.9568	0.1376	0.0740	0.0000	2026-11-04
36	aoxu aoxu1234	**14.9983**	34.9966	0.8874	0.7793	0.0243	2026-10-30
37	Javier Medel, MSc. iorideepp	**14.9989**	23.4832	1.1589	0.7191	0.0021	2026-10-15
38	World Reward thomasgomezz	**15.0255**	21.0174	0.1398	0.0798	0.0001	2026-11-01
39	SeanSiu2077 seansiu2077	**15.0399**	26.0803	0.1386	0.0734	0.0580	2026-11-03
40	EmAI Robotics heussers	**15.0668**	27.9991	1.3013	1.0012	0.0026	2026-11-04
41	sean2413 sean2413	**15.0977**	20.0174	0.9372	0.6505	0.0013	2026-11-01
42	deecod devutkarshpal	**15.1348**	89.0574	0.9558	2.0679	0.0003	2026-10-24
43	Null Walker yangxuyan	**15.1920**	22.4070	0.2620	0.2691	0.0097	2026-11-03
44	avinsingh avinsingh	**15.3011**	21.6181	0.1409	0.0762	0.0004	2026-11-03
45	TheTensorGeek wahidabdul	**15.4455**	84.2269	0.9720	0.7484	0.0003	2026-10-15
46	longbot imsyed	**15.4486**	24.2111	0.2748	0.2196	0.0033	2026-10-26
47	Tony tianny0078	**15.5170**	36.4004	0.1402	0.0743	0.0000	2026-10-23
48	Marcell marcellbalogh	**15.5628**	29.4734	0.1404	0.0725	0.0002	2026-11-03
—	CARI4D (commercial) Baseline * Baseline	15.5753	23.1625	1.9739	0.7262	0.0042	2026-10-01
49	fw095 fuquanwang	**15.5826**	39.0914	0.9666	0.1089	0.0014	2026-10-20
50	AI_CENTER ivanvoid	**15.7161**	24.8369	0.1409	0.0719	0.0046	2026-11-04
51	langsky qingqianlang	**15.8834**	33.3327	0.1397	0.0733	0.0003	2026-10-27
52	xiaopai7788 xiaopai7788	**15.9024**	49.1966	0.5010	0.5320	0.0012	2026-11-04
53	Zhanglele12138 zhanglele12138	**16.0893**	24.9156	0.1782	0.2226	0.0001	2026-10-28
54	MostlyKIGuess mostlykiguess	**16.3179**	55.9589	0.1686	0.0920	0.0138	2026-10-17
55	AIRC nhutoannguyen	**17.4192**	23.3842	0.2241	0.3000	0.0024	2026-10-30
56	Punya Malaiya punyamalaiya	**21.8727**	47.6085	0.1401	0.0946	0.0015	2026-10-31
57	Kelly To the H kellytotheh	**40.5981**	88.6178	0.1796	0.0836	0.0000	2026-11-04
58	Furqan Zaheer furqanzaheer	**73.6477**	169.9177	0.1453	0.0743	0.0000	2026-10-16
—	Liam Li hanboliamli no entry for CD-O, ACC-H, ACC-O, PEN	135.7567	-	-	-	-	2026-10-29

Updated Thu, 05 Nov 2026 01:43:17 GMT

Links to live track leaderboards on Kaggle CD-H CD-O ACC-H ACC-O PEN

* We use a commercial friendly recreation of CARI4D with retrained weights and some non-commercial modules replaced. Performance may differ slightly from the original implementation.

More Than a Leaderboard

REACH

INTEGRATION

AUTHORSHIP

Top teams are invited to present their solutions at our CoRL 2026 tutorial session.

Top solutions receive large-scale evaluation and are invited to upstream their solutions to a maintained code repository.

Teams are invited to co-author the post-challenge technical report, mapping where the field stands and setting the baseline for the next challenge.

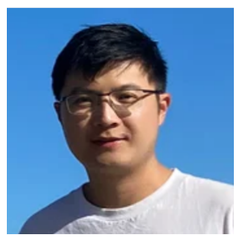
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Contact

For questions about the challenge, refer to our FAQ document or contact the organizers.

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